Does shifting the sample change DI? Equal versus unequal inclusion probabilities

12 × 12 fixed population · $N = 144$, $n = 8$ · all eight references independently fitted

Top row: shift (1, 1) | Bottom row: large shift (9, 8) | FIPs stay attached to population units

(a) Equal FIPs · original
DI = +0.198463015

Sample occupies 8/8 clusters

(b) Equal FIPs · shifted
DI = +0.198463015
ΔDI = +0.000000000; changed 0/144
Sample occupies 8/8 clusters

(c) Unequal FIPs · original
DI = +0.247754800

Sample occupies 7/8 clusters

(d) Unequal FIPs · shifted
DI = +0.247754800
ΔDI = +0.000000000; changed 0/144
Sample occupies 8/8 clusters

(e) Equal FIPs · original
DI = -0.997709500

Sample occupies 1/8 clusters

(f) Equal FIPs · shifted
DI = -0.997709500
ΔDI = +0.000000000; changed 0/144
Sample occupies 2/8 clusters

(g) Unequal FIPs · original
DI = -0.998149139

Sample occupies 1/8 clusters

(h) Unequal FIPs · shifted
DI = -0.998149139
ΔDI = +0.000000000; changed 0/144
Sample occupies 2/8 clusters

Dispersed sample Grid y

Concentrated sample Grid y

Grid x

Population unit; area ∝ FIP | Selected sample unit | Medoid | Medoid–sample match | Sample before shift

Equal FIPs: π = 1/18. Unequal FIPs: 0.0234375–0.0869141, summing to 8.

Colors show hard clusters and are aligned by overlap; unchanged colors are an observed result, not imposed reuse of the reference.